GinCol Lab · Aula

Student worksheet

Artistic Drawing I · 1st year of Bachillerato · Learning to look · Class 2 of 5 · With Marina

# Measuring with your pencil

Class video: https://youtu.be/naMO7Rhr7mk

The code leads to the video.

Name: _____ Date: _____

## Before you start

Have a pencil, a sheet of paper and something from your desk to hand: a bottle, a glass or a mug. Nothing else.

1. Look at the bottle or the mug in front of you, without measuring. How many times do you think its width fits into its height? Write it down.

2. **At the end of the class.** Measure it with the pencil, as the video shows, and compare it with what you wrote down.

## The gesture, step by step

Your pencil, with your arm outstretched, is a strip of Leonardo's glass: what you mark on it is what you see. What makes the measurement work is that **the distance from your eye to the pencil doesn't change** between one measurement and the next; the fully outstretched arm is the position that repeats itself.

1. Stay where you are and don't move. If you're sitting, don't move the chair.

2. Stretch your arm out fully, with your elbow straight. The pencil, straight, standing upright: don't tilt it towards you.

3. Close one eye, always the same one. If you can't close just one, cover it with your hand.

4. Put the tip of the pencil at one end of what you're measuring and slide your thumbnail down to the other. **Your measurement is the piece from the tip to your thumb**, not the whole pencil.

5. Without moving your thumb, take that measurement to another part and count how many times it fits.

6. On your sheet, decide the size: one line at the top and another at the bottom. Transfer the **proportion** to the paper, not the centimetres. Look first for where halfway falls.

7. Widths: lay the pencil flat, measure, stand it up without moving your thumb and compare it with the height.

8. The axis: let the pencil hang between two fingers, like a plumb line. It tells you what is directly above what.

9. Slanted lines aren't measured: they're copied. Lay the pencil along the line, read it like the hand of a clock and draw the same time on your sheet, with twelve at the top.

10. Very small things, by eye, within what you've already measured. And at the end, the line, light.

## While you watch the video

1. **Leonardo's advice.** What three things did Leonardo ask for to copy a landscape through a pane of glass? What gets marked on the glass?

2. **From the glass to the pencil.** Why is your pencil, with your arm outstretched, a strip of that glass? Which part of the pencil is your measurement?

3. **The Roman bottle.** Before measuring: do the mouth and the neck look shorter than the body to you? What do you get when you measure it? Where is the bottle's shoulder?

4. **Move closer.** If you move closer to the bottle, what changes on your pencil and what stays the same? What does it mean that the pencil measures proportions?

5. **Why still?** What happens to your measurement if you bend your elbow a little between one measurement and the next? Why?

6. **Why one eye?** What does the pencil do when you switch eyes? Why?

7. **Blocking in the bottle.** Put the steps in order: the line, the width, your spot, halfway, the size on the sheet, the axis, everything else.

8. **The first mistake.** Why aren't the pencil's centimetres transferred to the paper?

9. **Van Gogh.** How did he learn to draw? What did he write to his brother about books? In his 1888 street, what is under the ink?

## Your mug, half done

The earthenware mug in the video (Met, 12.221). Measure on the screen and block it in on your sheet.

| | **What you measure** | **What you get** |
|---|---|---|
| The width of the body, without the handle: how many times does it fit into the height? | | |
| Where does half the height fall? | | |
| How high up does the handle join at the bottom, counting from the foot? | | |
| **Check:** in your drawing, does its width fit one and a half times into its height? | | |

## The mistakes

Match each mistake with what it causes.

| **Mistake** | **What it causes** |
|---|---|
| 1. Transferring the pencil's centimetres to the paper | a. What you measure gets longer |
| 2. Bending your elbow between one measurement and the next | b. The pencil jumps in front of what you're measuring |

| Mistake | What it causes |
| --- | --- |
| 3. Switching eyes | c. The pencil looks bigger and the measurement no longer matches |
| 4. Tilting the pencil towards you | d. A tiny drawing, not the size you decided |
| 5. Measuring very small things | e. A small slip of the thumb is already a lot |

## Practice

1. Block in a bottle or a mug from your desk, measuring, and note in a corner which part you measured with, and where halfway falls.

2. **The Met's chair (66.10.5), the one in the video.** Draw it whole, measuring, or just its blocking in, in lines. Start with how many times its width, from the leg on the left to the leg on the right, fits into its height, from the top of the back to the lowest foot.

3. In the blocking in of the bottle that appears at the end of the video, someone bent their elbow when measuring one part. Say which one, and how you know.

## Keep thinking

If measuring helps a drawing look like its model, how come a heavily measured drawing can look stiff, and a slightly crooked one, alive? Think of drawings you like: are they measured? A clue: compare the road Van Gogh drew in 1881 with his 1888 street, and look for the pencil under the ink.

## For the whole topic

A couple of pencils, an ordinary one (HB) and a soft one (2B or 4B), an eraser and some sheets of paper. Your drawings are yours: you don't need to show them or upload them anywhere.

The quotations are from Leonardo da Vinci (*Treatise on Painting*, §87) and Vincent van Gogh (letter to his brother Theo, June 1881). The measuring hand and the student in profile are the channel's illustrations, made with AI. The bottle, the mug, the chair and the three works by Van Gogh belong to the Metropolitan Museum of Art in New York, which releases them free of rights (CC0).

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